

Robert Wang

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Highlights

- Computer science graduate with 4+ years experience in full stack web development and compiler design.
- Proficient knowledge in compiler theory and systematic engineering. Moderate experience with reinforcement learning algorithms and machine learning with Scikit-learn and PyTorch.
- Advanced knowledge in full stack development concepts, Very familiar with Next.js, Node.js, and Git, as well as backend concepts like Redis, Prisma, PostgreSQL, and MongoDB.
- Proficient with TypeScript, JavaScript, and Python. Competent experience with Assembly, C, C++, Java, and Haskell.

Education

UNIVERSITY OF CALIFORNIA, SANTA CRUZ

Santa Cruz, CA

Bachelor of Science, Computer Science

March 2025

Relevant coursework: Programming Language, Compiler Theory, Computer Systems Design, Computer Architecture, Software Development, Machine Learning, Data Structures and Algorithms, Algorithm Analysis, Computer Graphics

Professional Experience

Compiler Engineer Intern

Beijing, China

Extreme Intelligence Networking LTD.

December 2024 - Present

- Created a prototype transpiler for a custom programming language used for statistics tracking for the card game of Bridge.
- Implemented the lexical, syntactic, semantic analysis, as well as intermediate code generation functions for the program, transpiling the custom language into JavaScript code. Specific features I worked on are static variable mapping, keyword parsing, and intermediate code generation.
- Currently developing hand analysis functions. Market test is projected to begin between March and June, 2025.
- Transpiler and Dealer Demo: bridge.inl859.com

Full Stack Developer

Vancouver, Canada

BC 142434 (Tech Startup in Encrypted Video Communications)

December 2023 - February 2024

- Designed and built a responsive and efficient UI using React, TypeScript, Bootstrap, and WebRTC for a four-way encrypted video chat web application, enabling frictionless user video, camera, and mic control.
- Implemented Redis to streamline database architecture. Increased scalability and efficiency in logging, statistics keeping, system data recording and data persistence.
- Enabled language localization support for English and Simplified Chinese, allowing the company to expand into the Chinese-speaking market and adding hundreds of millions of potential users.
- Product URL: mslt.online

Marketing and Business Development Intern

Los Angeles, CA

ChargeFuze (IoT Startup in Consumer Goods Rental)

July 2021 - May 2023

- Collaborated with the software team to help incorporate Apple Pay to the company's payment system with Stripe APIs.
- Responsible for receiving customer feedbacks and driving continuous product improvements, including more user-friendly designs and broader payment methods. Cooperated with product team to implement improvements that resulted in 10% annual growth in revenue from music festivals and multi-year partnerships with Chase, Verizon, and Live Nation.
- Coordinated logistics, activations, and post-event load outs in California sporting and music events such as the Rose Bowl, Outside Lands, and BottleRock music festival, ensuring appealing product display and efficient operations.

Projects

AI's Impact on Hand Written Compilers

December 2024 - Present

- AI chatbots and LLMs are becoming more advanced on a daily basis, becoming an vital part in computer science. We can see their impact on the software development field, especially front-end. The goal of this project is to evaluate the impact of artificial intelligence's impact on compiler development, aiming to determine the accuracy of the models and whether they can increase efficiency significantly in compiler programming.
- Method: We are building a calculator program, starting from translating infix expressions to postfix and adding features step by step, such as support for variables, boolean and unary operators, and logarithmic and trigonometric functions. We will be using tools such as Google's Gemini Advanced to assist us during the development process. The usage, correctness, and quality of the results will be recorded and compared with the benefits of LLMs on software development.
- Progress: 25% completed, estimated finish date is end of March.